Module 10 Vocabulary

Algebraic Representation: A way to describe a transformation using equations that show how the coordinates of a figure change.

Algebraic Rule: A formula that describes how each point of a figure changes during a transformation.

Congruence: Congruence means that two shapes are exactly the same in size and shape. This means you could move one shape over the other, and they would fit perfectly on top of each other without stretching or compressing.

Congruent: Figures that have the same size and shape, with corresponding angles and sides equal.

Coordinate Plane: A two-dimensional surface formed by a horizontal line (x-axis) and a vertical line (y-axis) that intersect at the origin.

Dilation: Dilation is a transformation that changes the size of a shape but keeps its proportions the same. It either makes the shape larger or smaller using a specific point called the center of dilation and a multiplier known as the scale factor.

Image: The image is the new shape that you get after you have transformed the original shape (called the preimage) by moving, resizing, or rotating it. Think of it as the result or outcome of the transformation.

Non-Rigid Motion: A transformation, such as a dilation, that changes the size of a figure and does not preserve distance; it is used to justify similarity.

Origin: The point (0, 0) where the x-axis and y-axis meet in the coordinate plane.

Preimage: The preimage is the original shape before any transformations are applied to it. It's the starting point from which we create the image through transformations like translation, reflection, or dilation.

Reflection: A reflection is like placing a mirror along a line and flipping a shape over that line. The shape looks the same but is reversed, like seeing your mirror image.

Rigid Motion: A transformation that preserves size and shape, such as a reflection, rotation, or translation; it is used to justify congruence.

Rigid Transformation: A rigid transformation changes the position of a shape without altering its size or shape. Types of rigid transformations include translations, rotations, and

reflections. These transformations maintain the congruence between the preimage and the image.

Rotation: A rotation turns a shape around a fixed point, called the center of rotation, through a specific angle and direction (like clockwise or counterclockwise). The shape stays the same size and shape; it just spins around.

Scale Factor: The scale factor is a number that you multiply by the dimensions of the original shape (preimage) to get the dimensions of the new shape (image) during a dilation. A scale factor greater than 1 makes the shape larger, and a scale factor less than 1 makes it smaller.

Sequence of Transformations: A set of two or more transformations applied in a specific order to a figure.

Similarity: Two shapes are similar if they have the same shape but different sizes. This means they can be resized versions of each other, maintaining the proportions of their sides and angles. Similarity is achieved through transformations like dilation.

Transformation: A change in the position, size, or orientation of a figure on a plane.

Translation: A translation moves every point of a shape the same distance in the same direction. The shape looks exactly the same, just in a different place. It's like sliding a shape across the floor or shifting it up, down, left, or right.

Vector: A quantity with direction and magnitude that describes how far and in what direction to translate a figure.